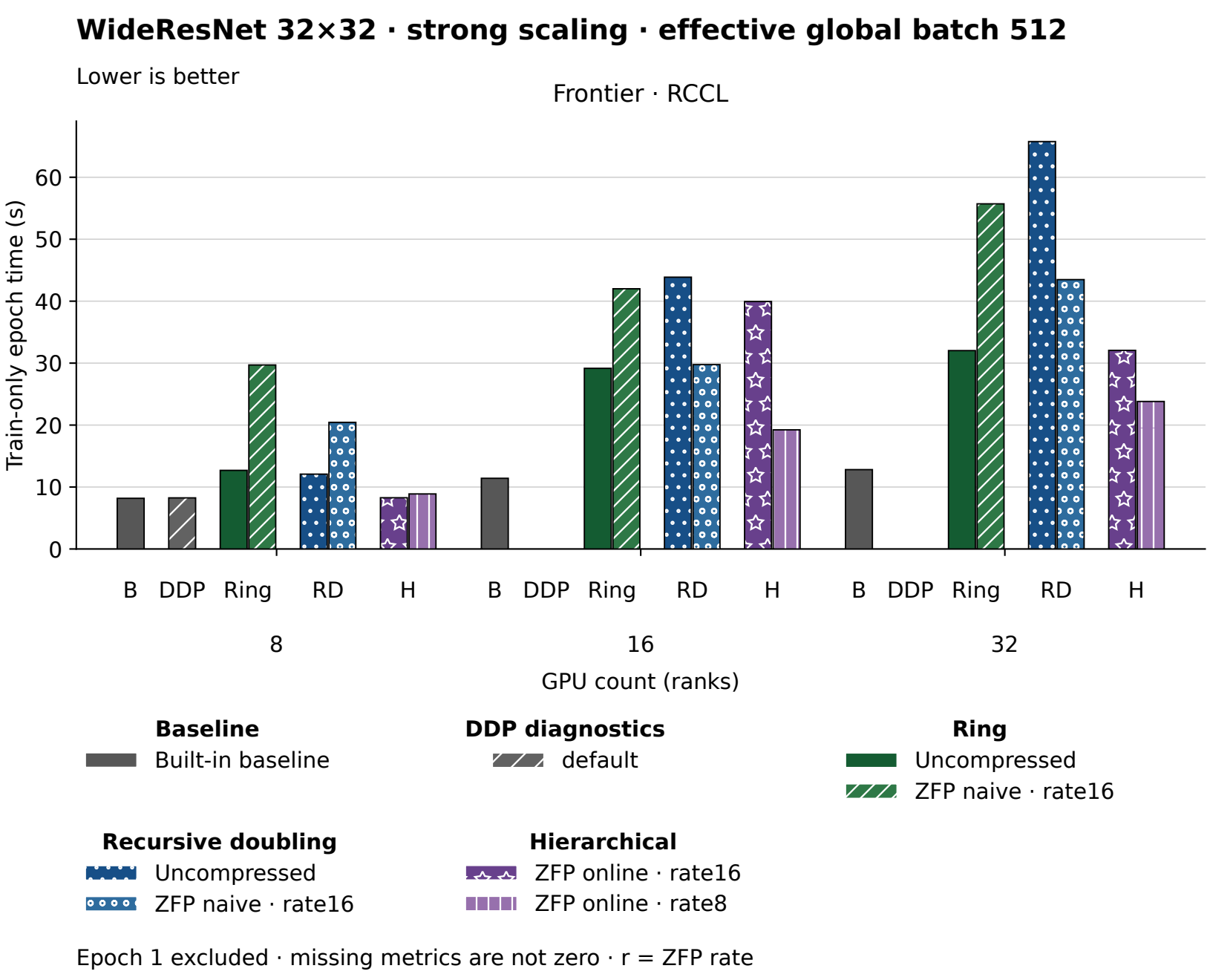

WideResNet 32×32 · strong scaling · effective global batch 512
Lower is better
Frontier · RCCL
Train-only epoch time (s)
0
10
20
30
40
50
60
B
DDP
Ring
RD
H
8
16
32
GPU count (ranks)
Baseline
Built-in baseline
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate